

JUNCHENG LU

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EDUCATION

Teachers College, Columbia University

M.A. in Cognitive Science in Education

September 2025 – Present

Beijing Normal University

B.S. in Psychology

September 2020 – July 2024

Selected coursework: Computational Cognitive Neuroscience; Brain Imaging Data Modeling; fMRI: Principles, Experiment Design and Data Analysis.

RESEARCH EXPERIENCE

Social Cognitive and Neural Sciences Lab

Research Assistant, Columbia University | PI: Dr. Jon Freeman

September 2025 – Present

Metacognition of speeded association-based decisions on the IAT

First author; manuscript in preparation

January 2026 – Present

- Quantified individual differences in the accuracy of decision-bias predictions using 2,986 IAT observations from 1,866 participants across four race- and age-based Implicit Association Tests.
- Dissociated latent processes underlying ingroup- versus outgroup-affirming response patterns by fitting multinomial processing tree (MPT) / quadruple process (Quad) models with maximum likelihood.
- Estimated hierarchical Bayesian DDM and Quad models to test which latent decision processes account for metacognitive accuracy across participants and tasks.

Computational modeling of speeded weapon-versus-tool decisions under racial face priming

September – December 2025

- Applied HDDM and MPT models to speeded weapon-versus-tool decisions under racial priming to examine how evidence accumulation and latent processing components contribute to biased decisions.
- Compared alternative cognitive architectures and related model-derived parameters to fMRI activation, evaluating mechanistic accounts of threat processing and cognitive control.

Intergroup Relation Neuroscience Lab

Research Assistant, Beijing Normal University | PI: Dr. Xiaochun Han

March 2022 – June 2025

Computational and Neural Mechanisms of Collateral-Harm Decisions

Revise and resubmit at Journal of Experimental Social Psychology; co-first author

March 2022 – June 2025

- Led the design, implementation, and data collection for two studies (total $N = 983$), integrating cross-cultural behavioral experiments in Chinese and U.S. samples with EEG.
- Conducted all behavioral, computational, and EEG analyses, integrating hierarchical Bayesian DDMs with EEGLAB-based ERP analyses to test how civilian race and pain expression modulate decision caution and its neural correlates.
- Observed a Race $\times$ Expression interaction in neural signals associated with decision caution despite no parallel condition differences in overt attack choices; contributed to manuscript drafting and revision with the PI.

Cognitive and neural mechanisms of discriminated-attack decisions in intergroup conflicts

Co-first author; manuscript in preparation

September 2023 – June 2025

- Analyzed behavioral and neural data from four studies testing how social-role information changes intergroup conflicts attack decisions, third-party judgments, and representations of warmth and competence.
- Fit logistic and linear mixed-effects models to estimate cross-cultural generalization, category-by-information interactions, and experimentally induced changes while accounting for participant- and role-level variation.
- Found that role information attenuated civilian–soldier differences across decision and representation measures; contributed to manuscript preparation.

MANUSCRIPTS

UNDER REVIEW

Lu, J.^{*}, Zhang, Y.^{*}, Han, X., Xiao, Z., & Han, X.[†] (2026). Computational and neural mechanisms of race-dependent influence of civilian suffering on collateral-harm decisions during intergroup conflicts. *Revise and resubmit, Journal of Experimental Social Psychology*.

IN PREPARATION

Zhang, Y.^{*}, Lu, J.^{*}, & Han, X.[†] (2026). Cognitive and neural mechanisms of discriminated-attack decisions in intergroup conflicts. *Manuscript in preparation*.

* Equal contribution. [†] Corresponding author.

CONFERENCE PRESENTATIONS

Lu, J., & Han, X.[†] (April 2024). Computational and neural mechanisms of decision-making causing same-race vs. other-race civilian casualties during intergroup conflicts. Selected for an oral presentation and poster presentation at the *Social and Affective Neuroscience Society*, Toronto, Canada. Unable to present due to visa restrictions.

TECHNICAL SKILLS

Programming	Python (HDDM, NumPy, pandas); R (brms, lme4, TreeBUGS, rtmpt); MATLAB (EEGLAB, SPM12, Psychtoolbox); JavaScript/HTML (jsPsych).
Computational Modeling	Hierarchical Bayesian inference; evidence-accumulation models (DDM/HDDM); process-dissociation models (MPT/Quad); GLMM/LMM; MCMC diagnostics; model comparison.
Neuroimaging	EEG/ERP preprocessing, time-locked analysis, and inference; fMRI preprocessing and modeling; model-based brain-behavior analysis.